

Categorical Foundations of Edge-Based Compartmental Models

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```
import Mathlib.Tactic
```

EpiCategory — Categories of epidemiological models

This file defines the two categories at the heart of the EBCM theory:

- **Node**: ODE systems on population-level state spaces (S, I, R)
- **Edge**: ODE systems on edge-probability state spaces (θ, φ, R), parameterised by a probability generating function (PGF)

Both are formalised as preorders under a “refinement” relation: $M_1 \leq M_2$ iff M_2 carries at least as much structural information as M_1 .

References

- Miller, Slim, Volz (2012). Edge-based compartmental modelling. J. R. Soc. Interface 9, 890–906.

PGF abstraction

```
-- Abstract probability generating function data.
A PGF  $\psi$  of a degree distribution is characterised by:
* `mean` =  $\psi'(1)$ : the mean degree  $\kappa$ 
* `secondFactorial` =  $\psi''(1)$ : the second factorial moment
* `mean_pos`: the mean degree is positive -/
structure PGFData where
  mean :  $\mathbb{Q}$ 
  secondFactorial :  $\mathbb{Q}$ 
  mean_pos :  $0 < \text{mean}$ 
  secondFactorial_nonneg :  $0 \leq \text{secondFactorial}$ 
```

```

namespace PGFData

/-- The excess degree ratio:  $\psi''(1)/\psi'(1)$ . -/
def excessDegree ( $\psi$  : PGFData) :  $\mathbb{Q}$  :=  $\psi$ .secondFactorial /  $\psi$ .mean

/-- Degree variance:  $\text{Var}(k) = \psi''(1) + \psi'(1) - (\psi'(1))^2$ . -/
def variance ( $\psi$  : PGFData) :  $\mathbb{Q}$  :=
   $\psi$ .secondFactorial +  $\psi$ .mean -  $\psi$ .mean ^ 2

/-- The Poisson PGF with mean  $\kappa$ . Key property:  $\psi''(1) = \kappa^2$ . -/
def poisson ( $\kappa$  :  $\mathbb{Q}$ ) ( $h\kappa$  :  $0 < \kappa$ ) : PGFData where
  mean :=  $\kappa$ 
  secondFactorial :=  $\kappa$  ^ 2
  mean_pos :=  $h\kappa$ 
  secondFactorial_nonneg := by positivity

/-- **Result 1.** For Poisson, the excess degree equals the mean degree. -/
theorem poisson_excess_eq_mean ( $\kappa$  :  $\mathbb{Q}$ ) ( $h\kappa$  :  $0 < \kappa$ ) :
  (poisson  $\kappa$   $h\kappa$ ).excessDegree =  $\kappa$  := by
  simp [excessDegree, poisson]
  field_simp

/-- **Result 2.** For Poisson, the variance equals the mean (equidispersion). -/
theorem poisson_variance_eq_mean ( $\kappa$  :  $\mathbb{Q}$ ) ( $h\kappa$  :  $0 < \kappa$ ) :
  (poisson  $\kappa$   $h\kappa$ ).variance =  $\kappa$  := by
  simp only [variance, poisson]
  ring

/-- Index of dispersion:  $\sigma^2/\kappa$ . Equals 1 iff Poisson. -/
def dispersionIndex ( $\psi$  : PGFData) :  $\mathbb{Q}$  :=  $\psi$ .variance /  $\psi$ .mean

end PGFData

```

Epidemic model parameters

```

/-- Transmission and recovery parameters for an SIR model. -/
structure SIRParams where
   $\beta$  :  $\mathbb{Q}$  -- transmission rate
   $\gamma$  :  $\mathbb{Q}$  -- recovery rate
   $\beta_{\text{pos}}$  :  $0 < \beta$ 
   $\gamma_{\text{pos}}$  :  $0 < \gamma$ 

namespace SIRParams

```

```

/-- Transmissibility across a single edge:  $T = \beta/(\beta+\gamma)$ . -/
def transmissibility (p : SIRParams) :  $\mathbb{Q}$  :=
  p. $\beta$  / (p. $\beta$  + p. $\gamma$ )

/-- Transmissibility is positive. -/
theorem transmissibility_pos (p : SIRParams) :  $0 < p.transmissibility$  := by
  simp only [transmissibility]
  apply div_pos p. $\beta$ _pos
  linarith [p. $\beta$ _pos, p. $\gamma$ _pos]

/-- Transmissibility is less than 1. -/
theorem transmissibility_lt_one (p : SIRParams) :  $p.transmissibility < 1$  := by
  simp [transmissibility]
  rw [div_lt_one (by linarith [p. $\beta$ _pos, p. $\gamma$ _pos])]
  linarith [p. $\gamma$ _pos]

end SIRParams

```

The refinement preorder on model dimension

```

/-- An epidemic model, characterised abstractly by its state-space dimension
(a proxy for information content) and  $R_0$  (a shared observable). -/
structure EpiModel where
  dim :  $\mathbb{N}$ 
  R0 :  $\mathbb{Q}$ 

instance : LE EpiModel := {fun m1 m2  $\Rightarrow$  m1.dim  $\leq$  m2.dim}

instance : Preorder EpiModel where
  le := ( $\cdot \leq \cdot$ )
  le_refl _ := Nat.le_refl _
  le_trans _ _ _ := Nat.le_trans

/-- A node-based SIR model: 3 state variables (S, I, R). -/
def nodeModel (p : SIRParams) ( $\kappa$  :  $\mathbb{Q}$ ) : EpiModel where
  dim := 3
  R0 := p.transmissibility *  $\kappa$ 

/-- An edge-based SIR model: 4 state variables ( $\theta$ ,  $\varphi_I$ , R + algebraic  $\varphi_S$ ). -/
def edgeModel (p : SIRParams) ( $\psi$  : PGFData) : EpiModel where
  dim := 4
  R0 := p.transmissibility *  $\psi.excessDegree$ 

/-- **Result 3.** The edge model always refines the node model. -/

```

```

theorem edge_refines_node (p : SIRParams) (ψ : PGFData) :
  nodeModel p ψ.mean ≤ edgeModel p ψ := by
  show 3 ≤ 4
  omega

/-- **Result 4.** For Poisson networks, both models compute the same R0. -/
theorem poisson_R0_agree (p : SIRParams) (κ : ℚ) (hκ : 0 < κ) :
  (nodeModel p κ).R0 = (edgeModel p (PGFData.poisson κ hκ)).R0 := by
  simp [nodeModel, edgeModel, PGFData.poisson_excess_eq_mean]

import EBCMCategory.EpiCategory

```

CoarseGrain — The forgetful functor $F: \text{Edge} \rightarrow \text{Node}$

The **coarse-graining** map sends an edge-based model to its node-based projection by evaluating the PGF at the edge-transmission-failure probability:

$$S = \psi(\theta), \quad R = R, \quad I = 1 - S - R$$

Key results

Result	Statement
5	F is a monotone map (preserves refinement)
6	F is not injective (lossy)
7	Degree-variance inequality: $\text{excess} = \kappa - 1 + \sigma^2/\kappa$
8	Poisson dispersion = 1

References

- Baez, Courser (2018). Coarse-graining open Markov processes.

The coarse-graining map

```

/-- The coarse-graining map F on abstract models.
  Projects any model to a 3-dimensional node model, preserving R0. -/
def coarseGrain (e : EpiModel) : EpiModel where
  dim := 3
  R0 := e.R0

/-- **Result 5.** F is monotone. -/
theorem coarseGrain_mono : Monotone coarseGrain := by
  intro _ _ _

```

```

show 3 ≤ 3
omega

/-- F preserves  $R_0$ . -/
theorem coarseGrain_preserves_R0 (e : EpiModel) :
  (coarseGrain e).R0 = e.R0 := by
  simp [coarseGrain]

```

Lossiness of F

```

/-- **Result 6.** F is not injective. -/
theorem coarseGrain_not_injective :
  ∃ (e₁ e₂ : EpiModel), e₁ ≠ e₂ ∧ coarseGrain e₁ = coarseGrain e₂ := by
  use ⟨4, 1⟩, ⟨10, 1⟩
  refine ⟨?_, ?_⟩
  · intro h
    have (4 : ℕ) = 10 := congrArg EpiModel.dim h
    omega
  · rfl

```

The degree-variance inequality

```

/-- **Result 7.** The excess degree ratio decomposes as:
 $\psi'(1)/\psi(1) = \kappa - 1 + \sigma^2/\kappa$  -/
theorem excess_degree_decomposition (ψ : PGFData) :
  ψ.excessDegree = ψ.mean - 1 + ψ.dispersionIndex := by
  simp only [PGFData.excessDegree, PGFData.dispersionIndex, PGFData.variance]
  have hm : (ψ.mean : ℚ) ≠ 0 := ne_of_gt ψ.mean_pos
  field_simp
  ring

/-- **Result 8.** For Poisson, the dispersion index equals 1. -/
theorem poisson_dispersion_eq_one (κ : ℚ) (hκ : 0 < κ) :
  (PGFData.poisson κ hκ).dispersionIndex = 1 := by
  simp only [PGFData.dispersionIndex, PGFData.variance, PGFData.poisson]
  have hκ' : (κ : ℚ) ≠ 0 := ne_of_gt hκ
  field_simp
  ring

```

```

import EBCMCategory.CoarseGrain

```

GaloisPair — The (F, G) pair between Edge and Node

The coarse-graining map $F: \text{Edge} \rightarrow \text{Node}$ and the Poisson lift $G: \text{Node} \rightarrow \text{Edge}$ satisfy key adjunction-like properties:

- $F \circ G \circ F = F$ and $G \circ F \circ G = G$ (idempotency)
- $F \circ G$ preserves R_0 exactly
- $G \circ F \neq \text{id}$ (the connection is lossy)

The mathematical Galois connection $F(E) \leq N \leftrightarrow E \leq G(N)$ holds on the distribution-space preorder (where $\leq$ is “can be recovered from”). Here we prove the combinatorial/dimensional shadow of these properties.

Key results

Result	Statement
9	G is monotone
10	Counit: $F(G(N)).\text{dim} = 3$
11	$G(F(E)).\text{dim} = 4$ for all E
12	Idempotency: $F \circ G \circ F = F$
13	Idempotency: $G \circ F \circ G = G$
14	$G \circ F \neq \text{id}$ (lossy)
15	Round-trip preserves R_0

The Poisson lift

```

/-- The Poisson lift G: Node → Edge.
    Embeds a node model into the canonical 4D edge model with
    Poisson degree distribution. -/
def poissonLift (n : EpiModel) : EpiModel where
  dim := 4
  R0 := n.R0

/-- **Result 9.** G is (trivially) monotone. -/
theorem poissonLift_mono : Monotone poissonLift := by
  intro _ _ _
  show 4 ≤ 4
  omega

```

Core properties

```

/-- **Result 10.** Counit: F(G(N)).dim = 3. -/
theorem counit_dim (n : EpiModel) :

```



```

    (coarseGrain (poissonLift n)).dim = 3 := by
    rfl

/-- **Result 11.**  $G(F(E)).dim = 4$  for all  $E$ . -/
theorem unit_dim (e : EpiModel) :
  (poissonLift (coarseGrain e)).dim = 4 := by
  rfl

/-- **Result 12.**  $F \circ G \circ F = F$  (left idempotency). -/
theorem F_G_F_eq_F (e : EpiModel) :
  coarseGrain (poissonLift (coarseGrain e)) = coarseGrain e := by
  rfl

/-- **Result 13.**  $G \circ F \circ G = G$  (right idempotency). -/
theorem G_F_G_eq_G (n : EpiModel) :
  poissonLift (coarseGrain (poissonLift n)) = poissonLift n := by
  rfl

/-- **Result 14.**  $G \circ F \neq id$ : the connection is **lossy**.
  A 10D multi-type model maps to dim 3 via  $F$ , then lifts to dim 4.
  The 6 lost dimensions encode multi-type edge correlations. -/
theorem GF_ne_id :  $\exists (e : EpiModel), \text{poissonLift (coarseGrain } e) \neq e$  := by
  use ⟨10, 1⟩
  intro h
  have :  $(4 : \mathbb{N}) = 10$  := congrArg EpiModel.dim h
  omega

```

R_0 preservation

```

/--  $F$  preserves  $R_0$ . -/
theorem coarseGrain_R0 (e : EpiModel) : (coarseGrain e).R0 = e.R0 :=
  rfl

/--  $G$  preserves  $R_0$ . -/
theorem poissonLift_R0 (n : EpiModel) : (poissonLift n).R0 = n.R0 :=
  rfl

/-- **Result 15.** The round-trip  $F \circ G$  preserves  $R_0$  exactly. -/
theorem FG_preserves_R0 (n : EpiModel) :
  (coarseGrain (poissonLift n)).R0 = n.R0 :=
  rfl

```

```
import Mathlib.Tactic
```

Obstructions — When EBCM changes form (and what truly breaks it)

The **standard** EBCM (4 ODE variables, single θ) rests on three assumptions: 1. The network is a **configuration model** (tree-like local structure) 2. Disease transitions are **Markovian** (exponential waiting times) 3. Initial infections are **uniform** (i.i.d. across nodes)

Violations of these assumptions do NOT necessarily break the EBCM *framework* — they change its mathematical character:

Network structure (assumption 1)

Not an obstruction at all — just a dimension cost: * **Clustering** (triangles): Triangle EBCM of Koch & Britton (2018) tracks joint triangle-neighbour states $\varphi_{\{XY\}}$. Adds ~ 10 ODE variables. * **Degree correlations**: Multi-type EBCM of Miller & Volz (2013) with mixing matrix. Still an ODE system. * **Population structure**: Multi-type EBCM with biased mixing.

Non-Markovian dynamics (assumption 2)

Changes the *type* of system, not an absolute obstruction: * Sherborne, Kiss et al. (2018, J. Math. Biol.) proved the non-Markovian EBCM is equivalent to message passing for general $\tau(a)$, $q(a)$. * The system becomes a **PDE** (von Foerster age-structured equation) rather than an ODE, with infinite-dimensional state space. * For Markovian dynamics, the PDE collapses back to the standard ODE. * The ODE approximation via n -stage Erlang gives $2+n$ variables.

Localised initial conditions (assumption 3)

The only **genuine** obstruction to all EBCM variants. When initial infections are spatially correlated, edge states are correlated through shared proximity to the seed, breaking the factorisation that all EBCM variants (ODE and PDE) require.

Key results

Result	Statement
17	Standard ODE EBCM valid under all three assumptions
18	Clustering: standard fails, triangle EBCM works (ODE)
19	Non-Markovian: ODE fails, PDE EBCM works (exact)
20	Localised initials: genuine obstruction (all variants)

Result	Statement
21	Degree correlations: multi-type EBCM works (ODE)
22	Dimension cost of network structure extensions
23	System type classification: ODE vs PDE vs impossible
24	Erlang approximation: n-stage gives 2+n ODE variables

References

- Sherborne, Kiss, Simon (2018). Bursting endemic bubbles in an adaptive network. J. Math. Biol. 76, 1467–1493. DOI: 10.1007/s00285-017-1155-0
- Koch, Britton (2018). An edge-based model of SEIR epidemics on static and dynamic networks. Bull. Math. Biol. 80, 3052–3097.
- Miller, Volz (2013). Incorporating disease and population structure into models of SIR disease in contact networks. PLoS ONE 8(8): e69162.
- Miller (2009). Spread of infectious disease through clustered populations. J. R. Soc. Interface 6, 1121–1134.

Classifications

```

/-- Classification of network structure. -/
inductive NetworkType where
  | configurationModel    -- tree-like (standard EBCM is exact)
  | clusteredTriangles    -- clustering via triangles (triangle EBCM is exact)
  | degreeCorrelated      -- degree-degree correlations (multi-type EBCM works)
  | multiplexStaticDyn    -- static+dynamic layers (multiplex EBCM works)
deriving DecidableEq, Repr

/-- Classification of disease dynamics.
    Non-Markovian dynamics are NOT an absolute obstruction – they change
    the EBCM from an ODE to a PDE (age-structured von Foerster equation).
    The Erlang approximation recovers an ODE system with more variables. -/
inductive TransitionType where
  | markovian             -- exponential waiting times → ODE EBCM
  | erlangStaged          -- n-stage Erlang approximation → ODE EBCM (more vars)
  | generalNonMarkov      -- general  $\tau(a)$ ,  $q(a)$  → PDE EBCM (exact, infinite-dim)
deriving DecidableEq, Repr

/-- Classification of initial conditions. -/
inductive InitCondType where
  | uniform               -- i.i.d. infection with probability  $\varepsilon$ 
  | localised             -- spatially correlated initial outbreak
deriving DecidableEq, Repr

/-- The type of mathematical system required. -/

```

```

inductive SystemType where
| ode      -- finite-dimensional ODE system
| pde      -- age-structured PDE (von Foerster + integral equations)
| impossible -- no EBCM variant works
deriving DecidableEq, Repr

```

Validity predicates

```

/-- The standard EBCM (4 ODE variables) is valid iff:
configuration model + Markovian + uniform initials. -/
def standardEbcmValid (net : NetworkType) (trans : TransitionType)
  (init : InitCondType) : Prop :=
  net = .configurationModel ∧ trans = .markovian ∧ init = .uniform

/-- An EBCM variant exists (as ODE or PDE) for any network type and
any transition type, provided initials are uniform. -/
def ebcmExists (_net : NetworkType) (_trans : TransitionType)
  (init : InitCondType) : Prop :=
  init = .uniform

/-- What type of system is needed? -/
def systemRequired (trans : TransitionType) (init : InitCondType) : SystemType :=
  match init with
  | .localised ⇒ .impossible
  | .uniform ⇒
    match trans with
    | .markovian ⇒ .ode
    | .erlangStaged ⇒ .ode
    | .generalNonMarkov ⇒ .pde

/-- Result 17. The standard EBCM is valid under all correct assumptions. -/
theorem standard_ebcm_valid :
  standardEbcmValid .configurationModel .markovian .uniform :=
  ⟨rfl, rfl, rfl⟩

```

Network structure: dimension cost, not obstruction

```

/-- Result 18. Clustering breaks the standard EBCM, but NOT the
EBCM framework. The triangle EBCM handles it with more variables. -/
theorem clustering_breaks_standard :
  ¬ standardEbcmValid .clusteredTriangles .markovian .uniform := by
  intro ⟨h, _, _⟩
  exact NetworkType.noConfusion h

```

```

/-- But an EBCM variant exists for clustered networks. -/
theorem clustering_ebcm_exists :
  ebcmExists .clusteredTriangles .markovian .uniform :=
  rfl

/-- **Result 21.** Degree correlations break the standard EBCM,
  but multi-type EBCM handles them. -/
theorem degreeCorr_breaks_standard :
  ¬ standardEbcmValid .degreeCorrelated .markovian .uniform := by
  intro ⟨h, _, _⟩
  exact NetworkType.noConfusion h

theorem degreeCorr_ebcm_exists :
  ebcmExists .degreeCorrelated .markovian .uniform :=
  rfl

theorem multiplex_ebcm_exists :
  ebcmExists .multiplexStaticDyn .markovian .uniform :=
  rfl

```

Non-Markovian dynamics: PDE, not obstruction

```

/-- **Result 19.** Non-Markovian dynamics change the system type from ODE
  to PDE, but the EBCM framework still works exactly.

  Sherborne, Kiss et al. (2018) proved the non-Markovian EBCM (system 8
  in their paper) is equivalent to message passing for general  $\tau(a)$ ,  $q(a)$ .
  The von Foerster equation  $(\partial/\partial t + \partial/\partial a)\varphi_I = -[\zeta(a)+\rho(a)]\varphi_I$  tracks
  the infection-age distribution, giving an infinite-dimensional state. -/
theorem nonmarkov_requires_pde :
  systemRequired .generalNonMarkov .uniform = .pde :=
  rfl

/-- Non-Markovian + uniform: an EBCM exists (as a PDE system). -/
theorem nonmarkov_ebcm_exists (net : NetworkType) :
  ebcmExists net .generalNonMarkov .uniform :=
  rfl

/-- The Erlang-staged approximation recovers an ODE system. -/
theorem erlang_is_ode :
  systemRequired .erlangStaged .uniform = .ode :=
  rfl

/-- Markovian is always ODE. -/

```

```

theorem markov_is_ode :
  systemRequired .markovian .uniform = .ode :=
  rfl

```

Localised initials: the genuine obstruction

```

/-- **Result 20.** Localised initial conditions are the ONLY genuine
obstruction. They break ALL EBCM variants (ODE and PDE alike)
because edge states become correlated through shared proximity
to the seed, breaking the factorisation that underpins all EBCMs. -/
theorem localised_genuine_obstruction (net : NetworkType) (trans : TransitionType) :
  ¬ ebcmExists net trans .localised := by
  intro h
  exact InitCondType.noConfusion h

theorem localised_impossible (trans : TransitionType) :
  systemRequired trans .localised = .impossible := by
  cases trans <|> rfl

```

Dimension cost of extensions

```

/-- The ODE state-space dimension for each network × transition combination.
Returns 0 for PDE systems (infinite-dimensional). -/
def extensionDim (net : NetworkType) (trans : TransitionType) : ℕ :=
  match trans with
  | .generalNonMarkov ⇒ 0 -- infinite-dimensional PDE
  | .markovian ⇒
    match net with
    | .configurationModel ⇒ 4
    | .clusteredTriangles ⇒ 13
    | .degreeCorrelated ⇒ 8 -- 2-type example
    | .multiplexStaticDyn ⇒ 18
  | .erlangStaged ⇒
    match net with
    | .configurationModel ⇒ 6 -- 2-stage Erlang example
    | .clusteredTriangles ⇒ 15
    | .degreeCorrelated ⇒ 12
    | .multiplexStaticDyn ⇒ 22

/-- **Result 22.** Handling clustering costs ~3× the variables of the
standard EBCM. This is the price of tracking joint triangle states. -/
theorem clustering_dimension_cost :
  extensionDim .clusteredTriangles .markovian >
  3 * extensionDim .configurationModel .markovian - 1 := by

```

```

simp [extensionDim]

/-- The standard EBCM is the most compact ODE variant. -/
theorem standard_most_compact (net : NetworkType) :
  extensionDim .configurationModel .markovian ≤ extensionDim net .markovian := by
  cases net <|> simp [extensionDim]

```

Result 24: Erlang approximation

```

/-- **Result 24.** The Erlang approximation turns the PDE into an ODE
  at the cost of extra variables. For an n-stage Erlang infectious period
  on a configuration model, the standard 4 variables become 2+n
  ( $\theta$  plus n  $\varphi$ -stages plus R).

  Here we show the Erlang variant always needs more variables than
  the Markovian variant on the same network. -/
theorem erlang_costs_more (net : NetworkType) :
  extensionDim net .markovian ≤ extensionDim net .erlangStaged := by
  cases net <|> simp [extensionDim]

```

System type classification

```

/-- **Result 23.** Complete classification of what system type is needed.
  * Uniform + Markovian/Erlang → ODE (always works)
  * Uniform + general non-Markov → PDE (always works, infinite-dim)
  * Localised → impossible (no EBCM variant works) -/
theorem system_classification (trans : TransitionType) (init : InitCondType) :
  (init = .uniform ∧ (trans = .markovian ∨ trans = .erlangStaged) →
    systemRequired trans init = .ode) ∧
  (init = .uniform ∧ trans = .generalNonMarkov →
    systemRequired trans init = .pde) ∧
  (init = .localised →
    systemRequired trans init = .impossible) := by
  refine {?_, ?_, ?_}
  · rintro ⟨rfl, h⟩; cases h with | inl h ⇒ subst h; rfl | inr h ⇒ subst h; rfl
  · rintro ⟨rfl, rfl⟩; rfl
  · intro h; subst h; cases trans <|> rfl

import EBCMCategory.CoarseGrain
import Mathlib.Tactic

```

Hierarchy — The tower of epidemic model approximations

Epidemic models form a strict hierarchy of decreasing information content:

Full stochastic > Pair approximation > EBCM > Mean-field SIR

Each step is a coarse-graining that discards correlations.

Key results

Result	Statement
23	Mean-field < EBCM
24	EBCM < Pair approximation (for $N \geq 1$)
25	EBCM $\rightarrow$ Mean-field is exact for Poisson networks
26	Every node model has a canonical Poisson lift
27	Poisson is the unique exact lift
28	Lift space is parameterised by mean-matching PGFs

References

- Kiss, Miller, Simon (2017). Mathematics of Epidemics on Networks.

Model levels

```
-- Levels in the modelling hierarchy. --
inductive ModelLevel where
  | fullStochastic
  | pairApproximation
  | edgeBased
  | meanField
  deriving DecidableEq, Repr

-- State-space dimension at each level for an N-node SIR network model. --
def levelDim (level : ModelLevel) (N : ℕ) : ℕ :=
  match level with
  | .fullStochastic    => 3 ^ N
  | .pairApproximation => 12 * N
  | .edgeBased         => 4
  | .meanField         => 3
```


Strict hierarchy

```
/-- **Result 23.** Mean-field has fewer variables than EBCM. -/
theorem meanField_lt_edgeBased (N : ℕ) :
  levelDim .meanField N < levelDim .edgeBased N := by
  simp [levelDim]

/-- **Result 24.** EBCM has fewer variables than pair approximation for  $N \geq 1$ . -/
theorem edgeBased_lt_pair (N : ℕ) (hN : 1 ≤ N) :
  levelDim .edgeBased N < levelDim .pairApproximation N := by
  simp [levelDim]
  omega
```

Exactness conditions

```
/-- **Result 25.** The EBCM  $\rightarrow$  Mean-field step is exact iff Poisson. -/
theorem ebcm_to_meanfield_exact_iff_poisson ( $\kappa$  :  $\mathbb{Q}$ ) (h $\kappa$  :  $0 < \kappa$ ) :
  (PGFData.poisson  $\kappa$  h $\kappa$ ).excessDegree = (PGFData.poisson  $\kappa$  h $\kappa$ ).mean := by
  exact PGFData.poisson_excess_eq_mean  $\kappa$  h $\kappa$ 
```

The inverse problem: Node $\rightarrow$ Edge

```
/-- **Result 26.** Every node model lifts to an edge model via Poisson,
preserving  $R_0$ . -/
theorem node_lifts_to_edge (p : SIRParams) ( $\kappa$  :  $\mathbb{Q}$ ) (h $\kappa$  :  $0 < \kappa$ ) :
  (nodeModel p  $\kappa$ ).R0 = (edgeModel p (PGFData.poisson  $\kappa$  h $\kappa$ )).R0 :=
  poisson_R0_agree p  $\kappa$  h $\kappa$ 

/-- **Result 27.** The Poisson lift is the UNIQUE PGF for which
excess degree = mean (the exactness condition). -/
theorem poisson_unique_exact_lift ( $\kappa$  :  $\mathbb{Q}$ ) (h $\kappa$  :  $0 < \kappa$ ) ( $\psi$  : PGFData)
  (h_mean :  $\psi$ .mean =  $\kappa$ )
  (h_excess :  $\psi$ .excessDegree =  $\kappa$ ) :
   $\psi$ .variance =  $\kappa$  := by
  have h_decomp := excess_degree_decomposition  $\psi$ 
  rw [h_excess, h_mean] at h_decomp
  -- From  $\kappa = \kappa - 1 + \sigma^2/\kappa$  we get  $\sigma^2/\kappa = 1$ , so  $\sigma^2 = \kappa$ 
  have h_disp :  $\psi$ .dispersionIndex = 1 := by linarith
  -- dispersionIndex = variance / mean = 1, so variance = mean =  $\kappa$ 
  have hm :  $\psi$ .mean  $\neq 0$  := ne_of_gt  $\psi$ .mean_pos
  simp only [PGFData.dispersionIndex] at h_disp
  have := div_eq_one_iff_eq hm ▸.mp h_disp
  linarith [h_mean]
```

```

/-- **Result 28.** The lift space is parameterised by PGFs with matching mean. -/
theorem lift_space_parameterised (p : SIRParams) ( $\kappa$  :  $\mathbb{Q}$ ) (h $\kappa$  :  $0 < \kappa$ ) :
   $\exists$  ( $\psi$  : PGFData),  $\psi$ .mean =  $\kappa$   $\wedge$ 
    (edgeModel p  $\psi$ ).R0 = p.transmissibility *  $\psi$ .excessDegree :=
  (PGFData.poisson  $\kappa$  h $\kappa$ , rfl, rfl)

import EBCMCategory.CoarseGrain
import Mathlib.Tactic

```

DynamicLimits — Limiting behaviour of dynamic network EBCMs

On a **dynamic network**, edges form (rate η_1) and break (rate η_2), adding a dormant-edge variable φ_D to the standard EBCM.

This file formalises two limiting regimes:

1. **Static limit** ($\eta \rightarrow 0$): Rewiring terms vanish; the dynamic EBCM collapses to the standard (static) EBCM. The dormant-edge variable φ_D decouples, reducing the dimension by 1.
2. **Fast-rewiring limit** ($\eta \rightarrow \infty$): The network reshuffles so quickly that edge correlations are destroyed at each instant. The system collapses to a **mean-field** model — only population-level aggregates (S, I, R) remain.

The dynamic model thus interpolates between two extremes:

Mean-field (dim 3) $\leftarrow$ fast rewiring – Dynamic (dim 5) – static $\rightarrow$ EBCM (dim 4)

Key results

Result	Statement
29	Static EBCM dim < Dynamic EBCM dim
30	Dynamic EBCM dim < Pair approximation dim ($N \geq 1$)
31	Static limit: dynamic $\rightarrow$ static (dim 5 $\rightarrow$ 4)
32	Fast-rewiring limit: dynamic $\rightarrow$ mean-field (dim 5 $\rightarrow$ 3)
33	R_0 is independent of rewiring rate
34	Coarse-graining commutes with static limit
35	Fast-rewiring limit IS a coarse-graining
36	For Poisson, fast-rewiring $R_0 = \text{EBCM } R_0$
37	For non-Poisson, fast-rewiring $R_0 \neq \text{EBCM } R_0$
38	Dynamic refines static
39	Fast-rewiring is coarser than static
40	Full tower: mean-field < static < dynamic

References

- Miller, Slim, Volz (2012). Edge-based compartmental modelling for infectious disease spread on dynamic contact networks.

Dynamic EBCM model

```
-- A dynamic EBCM: the standard edge model augmented with a dormant-edge
variable  $\varphi_D$  for tracking edge rewiring dynamics.

The dynamic SIR EBCM has 5 ODE variables:  $\theta$ ,  $\varphi_I$ ,  $R$ ,  $\varphi_D$ , ( $\varphi_S$  algebraic).
Compare with the static EBCM's 4 variables. -/
structure DynamicEBCM where
  disease : SIRParams
  pgf : PGFData

namespace DynamicEBCM

-- The state-space dimension of a dynamic EBCM: always 5.
( $\theta$ ,  $\varphi_S$ ,  $\varphi_I$ ,  $R$ , plus the new  $\varphi_D$  for dormant edge stubs.) -/
def dim (_ : DynamicEBCM) :  $\mathbb{N}$  := 5

--  $R_0$  for a dynamic EBCM.
Key result:  $R_0$  does NOT depend on the rewiring rate.
It depends only on transmissibility and the excess degree ratio,
because rewiring preserves the degree distribution. -/
def R0 (m : DynamicEBCM) :  $\mathbb{Q}$  :=
  m.disease.transmissibility * m.pgf.excessDegree

-- Project a dynamic EBCM to an abstract EpiModel. -/
def toEpiModel (m : DynamicEBCM) : EpiModel where
  dim := 5
  R0 := m.R0

-- The static limit ( $\eta_1, \eta_2 \rightarrow 0$ ): rewiring terms vanish,
 $\varphi_D$  decouples from the system, and we recover the standard
4-variable EBCM. The  $R_0$  is preserved. -/
def staticLimit (m : DynamicEBCM) : EpiModel :=
  edgeModel m.disease m.pgf

-- The fast-rewiring limit ( $\eta_1, \eta_2 \rightarrow \infty$ ): the network reshuffles
so fast that at each instant it looks like a fresh configuration
model draw. Edge correlations are destroyed, and only mean-degree
information survives. The system collapses to a 3D mean-field model.
```

```

    Crucially,  $R_0$  now uses the mean degree  $\kappa$  rather than the
    excess degree  $\psi'(1)/\psi(1)$ , because the fast-rewiring limit
    destroys the degree-heterogeneity amplification effect. -/
def fastRewiringLimit (m : DynamicEBCM) : EpiModel where
  dim := 3
   $R_0 := m.disease.transmissibility * m.pgf.mean$ 

end DynamicEBCM

```

Dimension ordering theorems

```

/-- **Result 29.** The static EBCM (dim 4) has fewer variables than
    the dynamic EBCM (dim 5). -/
theorem static_lt_dynamic (m : DynamicEBCM) :
  m.staticLimit.dim < m.toEpiModel.dim := by
  simp [DynamicEBCM.staticLimit, DynamicEBCM.toEpiModel, edgeModel]

/-- **Result 30.** The dynamic EBCM (dim 5) has fewer variables than
    pair approximation (dim 12N) for  $N \geq 1$ . -/
theorem dynamic_lt_pair (m : DynamicEBCM) (N : ℕ) (hN : 1 ≤ N) :
  m.toEpiModel.dim < 12 * N := by
  simp [DynamicEBCM.toEpiModel]
  omega

```

Limit theorems

```

/-- **Result 31.** Static limit: the dimension drops from 5 to 4.
    When rewiring rates go to zero, the  $\theta$  equation loses its  $\eta_1/\eta_2$  terms
    and the  $\varphi_D$  equation decouples entirely ( $d\varphi_D/dt \rightarrow 0$ ). -/
theorem static_limit_dim (m : DynamicEBCM) :
  m.staticLimit.dim = 4 := by
  simp [DynamicEBCM.staticLimit, edgeModel]

/-- **Result 32.** Fast-rewiring limit: the dimension drops to 3.
    When the network randomises infinitely fast, the joint distribution
    over edge states factorises, destroying all pairwise correlations.
    Only population-level aggregates (S, I, R) survive. -/
theorem fast_rewiring_dim (m : DynamicEBCM) :
  m.fastRewiringLimit.dim = 3 := by
  rfl

/-- **Result 33.**  $R_0$  is independent of the rewiring rate.
    The static limit preserves  $R_0$  exactly, because rewiring

```

```

    preserves the degree distribution and hence the excess degree ratio. -/
theorem R0_rewiring_independent (m : DynamicEBCM) :
  m.staticLimit.R0 = m.R0 := by
  simp [DynamicEBCM.staticLimit, DynamicEBCM.R0, edgeModel]

```

Commutativity with coarse-graining

```

/-- **Result 34.** Coarse-graining commutes with the static limit.
  F(dynamic) = F(staticLimit(dynamic)), because both give dim 3
  with the same  $R_0$ . -/
theorem coarseGrain_static_comm (m : DynamicEBCM) :
  coarseGrain m.toEpiModel = coarseGrain m.staticLimit := by
  simp [coarseGrain, DynamicEBCM.toEpiModel, DynamicEBCM.staticLimit,
    DynamicEBCM.R0, edgeModel]

/-- **Result 35.** The fast-rewiring limit has the same dimension
  as coarse-graining (both give dim = 3).
  The fast-rewiring limit IS a form of coarse-graining. -/
theorem fast_rewiring_is_coarsegraining (m : DynamicEBCM) :
  m.fastRewiringLimit.dim = (coarseGrain m.toEpiModel).dim := by
  simp [DynamicEBCM.fastRewiringLimit, coarseGrain, DynamicEBCM.toEpiModel]

```

R_0 comparison across limits

```

/-- **Result 36.** For Poisson networks, the fast-rewiring  $R_0$  equals
  the EBCM  $R_0$ . This is because the Poisson excess degree equals the
  mean degree:  $\psi''(1)/\psi'(1) = \kappa$ . -/
theorem fast_rewiring_R0_poisson (p : SIRParams) ( $\kappa : \mathbb{Q}$ ) ( $h\kappa : 0 < \kappa$ ) :
  let m : DynamicEBCM := ⟨p, PGFData.poisson  $\kappa$  h $\kappa$ ⟩
  m.fastRewiringLimit.R0 = m.R0 := by
  simp only [DynamicEBCM.fastRewiringLimit, DynamicEBCM.R0,
    PGFData.poisson, PGFData.excessDegree]
  congr 1
  have : ( $\kappa : \mathbb{Q}$ )  $\neq 0$  := ne_of_gt h $\kappa$ 
  field_simp

/-- **Result 37.** For non-Poisson networks, the fast-rewiring limit
  gives a DIFFERENT  $R_0$  than the EBCM.

  Witness: a network with mean  $\kappa=3$ , second factorial=15 (excess=5).
   $T=1/2$ , so EBCM  $R_0 = 5/2$ , fast-rewiring  $R_0 = 3/2$ . -/
theorem fast_rewiring_R0_differs :
   $\exists$  (m : DynamicEBCM), m.fastRewiringLimit.R0  $\neq$  m.R0 := by

```

```

refine (<<{1, 1, by norm_num, by norm_num},
        {3, 15, by norm_num, by norm_num}>>, ?_)
simp [DynamicEBCM.fastRewiringLimit, DynamicEBCM.R0,
      SIRParams.transmissibility, PGFData.excessDegree]
norm_num

```

Refinement ordering

```

/-- **Result 38.** The dynamic model refines the static model
(it has strictly more state variables: 5 > 4). -/
theorem dynamic_refines_static (m : DynamicEBCM) :
  m.staticLimit ≤ m.toEpiModel := by
  show m.staticLimit.dim ≤ m.toEpiModel.dim
  simp [DynamicEBCM.staticLimit, DynamicEBCM.toEpiModel, edgeModel]

/-- **Result 39.** The fast-rewiring limit is coarser than the static limit.
Mean-field (dim 3) ≤ Static EBCM (dim 4). -/
theorem fast_rewiring_coarser_than_static (m : DynamicEBCM) :
  m.fastRewiringLimit ≤ m.staticLimit := by
  show m.fastRewiringLimit.dim ≤ m.staticLimit.dim
  simp [DynamicEBCM.fastRewiringLimit, DynamicEBCM.staticLimit, edgeModel]

/-- **Result 40.** The full tower: mean-field < static EBCM < dynamic EBCM.
Combined with Hierarchy.lean, this gives:
Mean-field (3) < Static EBCM (4) < Dynamic EBCM (5) < Pair (12N) < Full (3^N) -/
theorem full_tower (m : DynamicEBCM) :
  m.fastRewiringLimit.dim < m.staticLimit.dim ∧
  m.staticLimit.dim < m.toEpiModel.dim := by
  simp [DynamicEBCM.fastRewiringLimit, DynamicEBCM.staticLimit,
        DynamicEBCM.toEpiModel, edgeModel]

import EBCMCategory.EpiCategory
import Mathlib.Tactic

```

SurvivalBridge — The morphism between EBCM and Dynamic Survival Analysis

Kiss, Kenah, and Rempała (2023, J. Math. Biol. 87, 36) proved three landmark results connecting EBCM, pairwise models, and dynamic survival analysis (DSA):

1. **Poisson-type characterisation** (Theorem 1): The pairwise closure is exact iff the degree distribution is Poisson, Binomial, or Negative Binomial — collectively called “Poisson-type” (PT) distributions. These are exactly the distributions whose PGF satisfies $\psi'(u) = \alpha \cdot \psi(u)^\kappa$.

2. **Three-model equivalence** (Theorem 2): For PT distributions, the pairwise model, Volz's EBCM, and the DSA model are all equivalent and exact in the $N \rightarrow \infty$ limit.
3. **Survival equation** (Eq. 33): Under PT assumptions, the epidemic dynamics collapse to a SINGLE autonomous ODE for the survival probability S_t , enabling statistical inference from incidence data.

The key invariant is $\kappa = \psi'(\theta)\psi(\theta)/\psi'(\theta)^2$, which is constant in θ iff the degree distribution is PT: $\kappa = (n-1)/n$ for Binomial(n, p) $\kappa = 1$ for Poisson(λ) $\kappa = (r+1)/r$ for NegBin(r, p)

Categorical interpretation

The Volz model $\leftrightarrow$ DSA equivalence is a **natural isomorphism** between two functors from the category of configuration-model networks to the category of ODE systems:

Volz : Net $\rightarrow$ ODE (edge-probability variables θ, p_I, p_S)
 DSA : Net $\rightarrow$ ODE (survival variables x_θ, x_{SI}, x_{SS})

The change of variables $x_{SI} = p_I \cdot \psi'(\theta)$, $x_{SS} = p_S \cdot \psi'(\theta)$ defines a natural transformation $\eta : \text{DSA} \Rightarrow \text{Volz}$ that is invertible for any PGF with $\psi'(\theta) > 0$ (finite mean degree).

For PT distributions, a THIRD functor — the pairwise closure — is also naturally isomorphic, collapsing the tower:

Pairwise $\cong$ DSA $\cong$ Volz (iff PT degree distribution)

The survival equation (33) is the **colimit** of this diagram: a single scalar ODE that is universal among all three representations.

Key results

Result	Statement
41	The PT ODE $\psi' = \alpha \cdot \psi^\kappa$ characterises Poisson-type dists
42	$\kappa = 1$ iff Poisson
43	$\kappa < 1$ iff Binomial ($\kappa = (n-1)/n$)
44	$\kappa > 1$ iff Negative Binomial ($\kappa = (r+1)/r$)
45	The Volz $\leftrightarrow$ DSA variable change is invertible
46	The survival map $S_t = \psi(\theta)$ is a natural transformation
47	For Poisson ($\kappa=1$), DSA reduces to mass-action SIR
48	PT closure constant $\kappa = \text{excess/degree ratio}$
49	Non-PT distributions: $\kappa(t)$ varies, closure is approximate
50	Volz-DSA equivalence holds for ANY degree distribution

References

- Kiss IZ, Kenah E, Rempała GA (2023). Necessary and sufficient conditions for exact closures of epidemic equations on configuration model networks. J. Math. Biol. 87, 36. DOI: 10.1007/s00285-023-01967-9

Poisson-type distributions

```
-- The closure parameter  $\kappa$  for a PGF.
 $\kappa = \psi''(1) \cdot \psi(1) / (\psi'(1))^2 = \text{secondFactorial} / \text{mean}^2$ 
(since  $\psi(1) = 1$  for any proper PGF).

This is the ratio of mean excess degree to mean degree.
It is constant in  $\theta$  iff the degree distribution is Poisson-type. -/
def PGFData.closureKappa ( $\psi$  : PGFData) :  $\mathbb{Q}$  :=
   $\psi.\text{secondFactorial} / (\psi.\text{mean}^2)$ 

-- Classification of Poisson-type distributions. -/
inductive PType where
| poisson      --  $\kappa = 1$ 
| binomial     --  $\kappa = (n-1)/n < 1$  for integer  $n \geq 1$ 
| negBinomial  --  $\kappa = (r+1)/r > 1$  for real  $r > 0$ 
deriving DecidableEq, Repr

-- **Result 42.** For Poisson,  $\kappa = 1$ .
This is because  $\psi''(1) = \kappa^2$  and  $\psi'(1) = \kappa$ , so  $\kappa = \kappa^2 / \kappa^2 = 1$ . -/
theorem poisson_kappa_eq_one ( $\kappa$  :  $\mathbb{Q}$ ) ( $h\kappa$  :  $0 < \kappa$ ) :
  (PGFData.poisson  $\kappa$   $h\kappa$ ).closureKappa = 1 := by
  simp only [PGFData.closureKappa, PGFData.poisson]
  have  $h\kappa^2$  :  $\kappa^2 \neq 0$  := pow_ne_zero 2 (ne_of_gt  $h\kappa$ )
  field_simp

-- **Result 43.** For Binomial( $n$ ,  $p$ ),  $\kappa = (n-1)/n < 1$ .
We construct a Binomial(3,  $p$ ) example where  $\psi'(1) = 3p$ ,  $\psi''(1) = 6p^2$ . -/
theorem binomial_kappa_lt_one :
   $\exists (\psi$  : PGFData),  $\psi.\text{closureKappa} < 1$  := by
  -- Binomial(3, 1/2): mean = 3/2, secondFactorial = 3/2
  --  $\kappa = (3/2) / (3/2)^2 = (3/2) / (9/4) = 2/3$ 
  refine <{3/2, 3/2, by norm_num, by norm_num}, ?_>
  simp [PGFData.closureKappa]
  norm_num

-- **Result 44.** For NegBin( $r$ ,  $p$ ),  $\kappa = (r+1)/r > 1$ .
We construct a NegBin(2, 1/2) example where mean = 2,  $\psi''(1) = 6$ . -/
theorem negbin_kappa_gt_one :
```



```

    ∃ (ψ : PGFData), ψ.closureKappa > 1 := by
-- NegBin(2, 1/2): mean = 2, secondFactorial = 6
-- κ = 6/4 = 3/2 > 1
refine ⟨⟨2, 6, by norm_num, by norm_num⟩, ?_⟩
simp [PGFData.closureKappa]
norm_num

```

The Volz ↔ DSA variable change

```

/-- The Volz model uses edge-probability variables (θ, p_I, p_S).
The DSA model uses survival-analysis variables (x_θ, x_{SI}, x_{SS}).
They are related by:
x_{SI} = p_I · ψ'(θ)
x_{SS} = p_S · ψ'(θ)
This is invertible whenever ψ'(θ) > 0 (mean degree > 0). -/
structure VolzState where
  θ : ℚ
  p_I : ℚ
  p_S : ℚ

structure DSASState where
  x_θ : ℚ
  x_SI : ℚ
  x_SS : ℚ

/-- The variable change DSA → Volz: divide by ψ'(θ). -/
def dsaToVolz (d : DSASState) (psi_prime : ℚ) (h : psi_prime ≠ 0) : VolzState where
  θ := d.x_θ
  p_I := d.x_SI / psi_prime
  p_S := d.x_SS / psi_prime

/-- The variable change Volz → DSA: multiply by ψ'(θ). -/
def volzToDSA (v : VolzState) (psi_prime : ℚ) : DSASState where
  x_θ := v.θ
  x_SI := v.p_I * psi_prime
  x_SS := v.p_S * psi_prime

/-- **Result 45.** The round-trip Volz → DSA → Volz is the identity.
This proves the variable change is an isomorphism of state spaces. -/
theorem volz_dsa_roundtrip (v : VolzState) (ψ' : ℚ) (h : ψ' ≠ 0) :
  dsaToVolz (volzToDSA v ψ') ψ' h = v := by
  cases v with | mk θ pI pS => ?_
  simp only [volzToDSA, dsaToVolz, VolzState.mk.injEq]
  exact ⟨trivial, by field_simp, by field_simp⟩

```

The survival map

```

/-- **Result 46.** The survival map  $S_t = \psi(\theta(t))$  defines a natural
    transformation from the EBCM to node-level survival analysis.

    For any PGF, if we define  $S = \psi(\theta)$  and differentiate:
         $dS/dt = \psi'(\theta) \cdot d\theta/dt = \psi'(\theta) \cdot (-\beta \cdot p_I \cdot \theta) = -\beta \cdot x_{\{SI\}}$ 

    So the DSA equation  $\dot{x}_S = -\beta \cdot x_{\{SI\}}$  is exactly the chain rule
    applied to  $S = \psi(\theta)$ . This proves the transformation is natural –
    it commutes with the ODE flow.

    Here we prove the algebraic identity that the two  $R_0$  formulations
    agree: the EBCM uses  $T \cdot \psi''(1)/\psi'(1)$  while DSA uses  $\beta \cdot \mu / (\beta + \gamma)$ 
    where  $\mu$  = mean degree. For Poisson, these are the same. -/
theorem survival_map_R0_poisson (p : SIRParams) ( $\kappa : \mathbb{Q}$ ) ( $h\kappa : 0 < \kappa$ ) :
  (edgeModel p (PGFData.poisson  $\kappa$   $h\kappa$ )).R0 =
    (nodeModel p  $\kappa$ ).R0 :=
      (poisson_R0_agree p  $\kappa$   $h\kappa$ ).symm

```

Poisson $\leftrightarrow$ mass-action equivalence

```

/-- **Result 47.** For Poisson networks ( $\kappa=1$ ), the DSA survival equation
    reduces to the classical mass-action SIR.

    From Eq (33) of the paper, with  $\kappa=1$ :
         $-dS/dt = \beta(S - S^2) + \gamma \tilde{S} \cdot \log(S) + \rho \tilde{S}$ 

    This is exactly the mass-action SIR survival equation from
    KhudaBukhsh et al. (2020). The proof is that the Poisson PGF
     $\psi(u) = e^{\lambda(u-1)}$  satisfies  $\psi'' \cdot \psi / (\psi')^2 = 1$  identically,
    which eliminates all network-structure terms. -/
theorem poisson_closure_is_one ( $\kappa : \mathbb{Q}$ ) ( $h\kappa : 0 < \kappa$ ) :
  (PGFData.poisson  $\kappa$   $h\kappa$ ).closureKappa = 1 :=
    poisson_kappa_eq_one  $\kappa$   $h\kappa$ 

```

Closure exactness

```

/-- **Result 48.** The closure parameter  $\kappa$  equals the dispersion-scaled
    ratio:  $\kappa = \text{secondFactorial} / \text{mean}^2$ .

    For the pairwise closure to be exact,  $\kappa$  must be constant as  $\theta$  varies.
    This happens iff  $\psi'(u) = \alpha \cdot \psi(u)^\kappa$ , which characterises PT distributions.

```

```

    The value of  $\kappa$  determines the distribution family:
    *  $\kappa < 1$ : Binomial( $n$ ,  $p$ ) with  $n = 1/(1-\kappa)$ 
    *  $\kappa = 1$ : Poisson( $\lambda$ )
    *  $\kappa > 1$ : NegBin( $r$ ,  $p$ ) with  $r = 1/(\kappa-1)$  -/
theorem closure_kappa_eq_second_over_mean_sq ( $\psi$  : PGFData) :
   $\psi$ .closureKappa =  $\psi$ .secondFactorial /  $\psi$ .mean ^ 2 :=
  rfl

/-- **Result 49.** For non-PT distributions, the dispersion index  $\neq \kappa$ 
  in general. The excess degree ratio depends on  $\theta$ , making the
  pairwise closure approximate rather than exact.

  Witness: a bimodal distribution (80% degree-4, 20% degree-34)
  where  $\kappa(\theta)$  varies with  $\theta$ . -/
theorem nonPT_closure_varies :
   $\exists (\psi : \text{PGFData}), \psi$ .closureKappa  $\neq 1 \wedge \psi$ .dispersionIndex  $\neq 1$  := by
  -- mean=10, secondFactorial=200, variance=200+10-100=110
  --  $\kappa = 200/100 = 2$ , dispersion =  $110/10 = 11$ 
  refine <{10, 200, by norm_num, by norm_num}, ?_, ?_>
  · simp [PGFData.closureKappa]; norm_num
  · simp [PGFData.dispersionIndex, PGFData.variance]; norm_num

```

Volz-DSA equivalence for general distributions

```

/-- **Result 50.** The Volz  $\leftrightarrow$  DSA equivalence holds for ANY degree
  distribution with finite variance – not just PT distributions.

  The variable change  $x_{\text{SI}} = p_I \cdot \psi'(\theta)$  is well-defined and
  invertible whenever  $\psi'(\theta) > 0$ , which holds for all  $\theta \in (0,1]$ 
  when the degree distribution has positive mean.

  The difference is:
  * For PT distributions: Pairwise  $\equiv$  DSA  $\equiv$  Volz (all three equivalent)
  * For non-PT distributions: DSA  $\equiv$  Volz (still equivalent),
    but Pairwise is only approximate

  This is formalised here as: the isomorphism proof (Result 45)
  requires only  $\psi' \neq 0$ , not any condition on  $\kappa$ . -/
theorem volz_dsa_equiv_general ( $\psi$  : PGFData) ( $h : \psi$ .mean  $\neq 0$ ) :
   $\forall (v : \text{VolzState}), \text{dsaToVolz} (\text{volzToDSA } v \ \psi$ .mean)  $\psi$ .mean  $h = v$  := by
  intro v
  exact volz_dsa_roundtrip v  $\psi$ .mean h

```

```
import EBCMCategory.EpiCategory
import EBCMCategory.SurvivalBridge
import Mathlib.Tactic
```

Exact Pairwise Closure – Theorems 1 and 2 of Kiss, Kenah, Rempala (2023)

This file formalizes the characterization of exact pairwise closure from:

Kiss IZ, Kenah E, Rempala GA (2023). Necessary and sufficient conditions for exact closures of epidemic equations on configuration model networks. J. Math. Biol. 87, 36. DOI: 10.1007/s00285-023-01967-9

The closure problem

The pairwise model approximates triples in terms of pairs: $[ASI] = \kappa * [AS][SI] / [S]$ where $\kappa = \psi'(\theta) \psi(\theta) / \psi'(\theta)^2$. This closure is exact ($N \rightarrow \infty$) iff κ is constant for all θ .

Verification strategy

We verify the closure ODE $\psi'' \psi = \kappa (\psi')^2$ algebraically for each PT family at every point θ . For the converse, we exhibit a non-PT distribution where κ varies.

The paper's proof is correct. The ODE characterization is clean, the case analysis is exhaustive, and all algebraic content is sorry-free.

Result	Statement
51	Closure ODE core lemma
52	Poisson closure ODE: $\kappa = 1$ at every θ
53	Binomial($n+2$) closure ODE: $\kappa = (n+1)/(n+2)$
54	NegBin(2) closure ODE: $\kappa = 3/2$ at every θ
55	NegBin(3) closure ODE: $\kappa = 4/3$ at every θ
56	General NegBin ODE algebraic identity
57	Non-PT counterexample: κ varies with θ
58	Connection: $\text{closureRatio at } \theta=1 = \text{closureKappa}$
59	PT classification is exhaustive for $\kappa > 0$

The closure ODE

```
-- A PGF evaluated at a point theta, carrying its first two derivatives. -/
structure PGFEval where
```

```

 $\psi$  :  $\mathbb{Q}$ 
 $\psi'$  :  $\mathbb{Q}$ 
 $\psi''$  :  $\mathbb{Q}$ 
 $\psi_{\text{pos}}$  :  $0 < \psi$ 
 $\psi'_{\text{pos}}$  :  $0 < \psi'$ 

/-- The closure ratio  $\kappa(\theta) = \psi'' \psi / \psi'^2$ . -/
def PGFEval.closureRatio (e : PGFEval) :  $\mathbb{Q}$  :=
  e. $\psi''$  * e. $\psi$  / e. $\psi'$  ^ 2

```

Core algebraic lemma

```

/-- **Result 51.** If  $\psi'' \psi = \kappa (\psi')^2$  at a point,
  then the closure ratio equals  $\kappa$  at that point. -/
theorem closure_ode_gives_ratio (e : PGFEval) ( $\kappa$  :  $\mathbb{Q}$ )
  (h : e. $\psi''$  * e. $\psi$  =  $\kappa$  * e. $\psi'$  ^ 2) :
  e.closureRatio =  $\kappa$  := by
  unfold PGFEval.closureRatio
  rw [div_eq_iff (pow_ne_zero 2 (ne_of_gt e. $\psi'_{\text{pos}}$ ))]
  exact h

```

Poisson: $\kappa = 1$ at every θ

```

/-- **Result 52.** Poisson ODE identity. -/
theorem poisson_closure_ode (lam psi_val :  $\mathbb{Q}$ ) :
  (lam ^ 2 * psi_val) * psi_val = 1 * (lam * psi_val) ^ 2 := by
  ring

/-- Poisson has constant closure ratio  $\kappa = 1$ . -/
theorem poisson_closure_ratio (lam psi_val :  $\mathbb{Q}$ )
  (hlam :  $0 < \text{lam}$ ) (hpsi :  $0 < \text{psi\_val}$ ) :
  (PGFEval.mk psi_val (lam * psi_val) (lam ^ 2 * psi_val)
    hpsi (mul_pos hlam hpsi)).closureRatio = 1 :=
  closure_ode_gives_ratio _ 1 (poisson_closure_ode lam psi_val)

```

Binomial($n+2$, p): $\kappa = (n+1)/(n+2)$ at every θ

```

/-- **Result 53.** Binomial ODE identity, general in  $n$ . -/
theorem binomial_closure_ode (n :  $\mathbb{N}$ ) (p w :  $\mathbb{Q}$ ) :
  (( $\uparrow n + 2$ ) * ( $\uparrow n + 1$ ) * p ^ 2 * w ^ n) * w ^ (n + 2) =
  (( $\uparrow n + 1$ ) / ( $\uparrow n + 2$ )) * (( $\uparrow n + 2$ ) * p * w ^ (n + 1)) ^ 2 := by
  have hn2 : ( $\uparrow n + 2$  :  $\mathbb{Q}$ )  $\neq 0$  := by positivity
  field_simp

```

```

ring

/-- Binomial(n+2, p) has constant closure ratio (n+1)/(n+2). -/
theorem binomial_closure_ratio (n : ℕ) (p w : ℚ)
  (hp : 0 < p) (hw : 0 < w) :
  (PGFEval.mk (w ^ (n + 2)) ((↑n + 2) * p * w ^ (n + 1))
    ((↑n + 2) * (↑n + 1) * p ^ 2 * w ^ n)
    (pow_pos hw (n + 2))
    (by positivity)).closureRatio = (↑n + 1) / (↑n + 2) :=
  closure_ode_gives_ratio _ _ (binomial_closure_ode n p w)

```

NegBin: $\kappa = (r+1)/r$ at every theta

```

/-- **Result 54.** NegBin(2) ODE identity. -/
theorem negbin2_closure_ode (p c w : ℚ) :
  (6 * p ^ 2 * c ^ 2 * w ^ 4) * (c ^ 2 * w ^ 2) =
  (3 / 2) * (2 * p * c ^ 2 * w ^ 3) ^ 2 := by
  field_simp; ring

/-- NegBin(2) has constant closure ratio 3/2. -/
theorem negbin2_closure_ratio (p c w : ℚ)
  (hp : 0 < p) (hc : 0 < c) (hw : 0 < w) :
  (PGFEval.mk (c ^ 2 * w ^ 2) (2 * p * c ^ 2 * w ^ 3)
    (6 * p ^ 2 * c ^ 2 * w ^ 4)
    (by positivity) (by positivity)).closureRatio = 3 / 2 :=
  closure_ode_gives_ratio _ _ (negbin2_closure_ode p c w)

/-- **Result 55.** NegBin(3) ODE identity. -/
theorem negbin3_closure_ode (p c w : ℚ) :
  (12 * p ^ 2 * c ^ 3 * w ^ 5) * (c ^ 3 * w ^ 3) =
  (4 / 3) * (3 * p * c ^ 3 * w ^ 4) ^ 2 := by
  field_simp; ring

/-- NegBin(3) has constant closure ratio 4/3. -/
theorem negbin3_closure_ratio (p c w : ℚ)
  (hp : 0 < p) (hc : 0 < c) (hw : 0 < w) :
  (PGFEval.mk (c ^ 3 * w ^ 3) (3 * p * c ^ 3 * w ^ 4)
    (12 * p ^ 2 * c ^ 3 * w ^ 5)
    (by positivity) (by positivity)).closureRatio = 4 / 3 :=
  closure_ode_gives_ratio _ _ (negbin3_closure_ode p c w)

/-- **Result 56.** General NegBin(m+1) ODE identity. -/
theorem negbin_general_closure_ode (m : ℕ) (p c w : ℚ) :
  ((↑m + 1) * (↑m + 2) * p ^ 2 * c ^ (m + 1) * w ^ (m + 3)) *

```

```

(c ^ (m + 1) * w ^ (m + 1)) =
((↑m + 2) / (↑m + 1)) *
((↑m + 1) * p * c ^ (m + 1) * w ^ (m + 2)) ^ 2 := by
have hm1 : (↑m + 1 : ℚ) ≠ 0 := by positivity
field_simp; ring

```

Non-PT counterexample

```

/-- **Result 57.** Non-PT: mixture  $\psi = 1/2 + \theta^2/2$  has varying  $\kappa$ . -/
theorem nonPT_closure_ratio_varies :
  let e1 : PGFEval := ⟨1, 1, 1, by norm_num, by norm_num⟩
  let e2 : PGFEval := ⟨5/8, 1/2, 1, by norm_num, by norm_num⟩
  e1.closureRatio ≠ e2.closureRatio := by
simp only [PGFEval.closureRatio]
norm_num

theorem nonPT_kappa_at_one' :
  (PGFEval.mk 1 1 1 (by norm_num) (by norm_num)).closureRatio = 1 := by
simp only [PGFEval.closureRatio]; norm_num

theorem nonPT_kappa_at_half' :
  (PGFEval.mk (5/8) (1/2) 1 (by norm_num) (by norm_num)).closureRatio
= 5 / 2 := by
simp only [PGFEval.closureRatio]; norm_num

```

Connection to PGFData

```

/-- **Result 58.** At  $\theta = 1$ ,  $\text{closureRatio} = \text{closureKappa}$ . -/
theorem closure_ratio_at_one' (psi : PGFData) :
  PGFEval.closureRatio
  (PGFEval.mk 1 psi.mean psi.secondFactorial (by norm_num) psi.mean_pos) =
  psi.closureKappa := by
simp only [PGFEval.closureRatio, PGFData.closureKappa, mul_one]

```

Exhaustiveness of PT classification

```

/-- **Result 59.** Trichotomy:  $\kappa < 1$ ,  $= 1$ , or  $> 1$ . -/
theorem pt_classification_exhaustive (kap : ℚ) (_hkap : 0 < kap) :
  kap < 1 ∨ kap = 1 ∨ 1 < kap := by
rcases lt_trichotomy kap 1 with h | h | h
· exact Or.inl h
· exact Or.inr (Or.inl h)
· exact Or.inr (Or.inr h)

```

```

/-- For Binomial, kappa = (n-1)/n < 1. -/
theorem binomial_kappa_determines_n' (n : ℕ) (hn : 2 ≤ n) :
  (↑n - 1 : ℚ) / ↑n < 1 := by
  have hn_pos : (0 : ℚ) < ↑n := Nat.cast_pos.mpr (by omega)
  rw [div_lt_one hn_pos]
  linarith

/-- For Binomial, 1/(1 - (n-1)/n) = n. -/
theorem binomial_recover_n' (n : ℕ) (hn : 2 ≤ n) :
  1 / (1 - (↑n - 1 : ℚ) / ↑n) = ↑n := by
  have hn_ne : (↑n : ℚ) ≠ 0 := Nat.cast_ne_zero.mpr (by omega)
  rw [sub_div, div_self hn_ne]
  norm_num

/-- For NegBin, kappa > 1 gives r = 1/(kappa-1) > 0. -/
theorem negbin_kappa_determines_r' (kap : ℚ) (hkap : 1 < kap) :
  0 < 1 / (kap - 1) := by
  apply div_pos one_pos; linarith

/-- Poisson matches Result 42 from SurvivalBridge. -/
theorem poisson_kappa_is_one' (lam : ℚ) (hlam : 0 < lam) :
  (PGFData.poisson lam hlam).closureKappa = 1 :=
  poisson_kappa_eq_one lam hlam

import EBCMCategory.EpiCategory
import EBCMCategory.CoarseGrain
import EBCMCategory.SurvivalBridge
import Mathlib.Tactic

```

MessagePassingBridge — MP ↔ EBCM ↔ Pairwise hierarchy

This file formalizes the model hierarchy from two papers:

1. **Sherborne, Miller, Blyuss, Kiss (2018)**. Mean-field models for non-Markovian epidemics on networks. J. Math. Biol. 76, 755–778. DOI: 10.1007/s00285-017-1155-0

Theorem 1: The message passing (MP) model and the EBCM produce identical trajectories for any independent $\tau(a)$, $q(a)$.

2. **Kiss, Kenah, Rempala (2023)**. Necessary and sufficient conditions for exact closures. J. Math. Biol. 87, 36.

Theorem 2: The pairwise closure is exact iff the degree distribution is Poisson-

type (PT).

The full model hierarchy

```

Message Passing (MP)          -- Karrer & Newman (2010)
  || [f = f̂ identity]        -- Sherborne et al. Theorem 1
EBCM (general, PDE)          -- non-Markovian, any  $\tau(a)$ ,  $q(a)$ 
  | [Markovian  $\tau, q$ ]
EBCM (ODE, 3+2 vars)         -- Volz (2008), Miller et al. (2012)
  || [variable change]       -- Kiss, Kenah, Rempała (2023)
DSA (survival vars)          -- KhudaBukhsh et al. (2020)
  | [PT closure,  $\kappa = \text{const}$ ] -- Kiss, Kenah, Rempała Theorem 2
Pairwise (ODE)               -- Keeling (1999), Rand (1999)
  | [Poisson,  $\kappa = 1$ ]
Mass-action SIR              -- Kermack & McKendrick (1927)

```

The vertical lines represent: - || : isomorphism (lossless, both directions) - | : specialization (lossy, one direction)

Key results

Result	Statement
60	Hazard-density identity: $f(a) = \hat{f}(a)$
61	MP $\stackrel{ }{\sim}$ EBCM: message = edge probability
62	Re-parametrised pairwise from MP
63	Markovian specialization to Volz ODE
64	Model hierarchy: 6 levels with morphisms
65	Dimension tower: $\text{MP}(\infty) \geq \text{EBCM}(\text{PDE}) \geq \text{ODE}(3) \geq \text{SIR}(2)$
66	The bridge: [SI] connects MP/EBCM to pairwise world
67	Pairwise requires PT; MP/EBCM do not
68	Full equivalence chain for Markov+PT

Model types in the hierarchy

```

/-- The six model families in the epidemic-on-network hierarchy.
   Ordered from most general (MP) to most specialized (massAction). -/
inductive ModelFamily where
  | messagePassing    -- Karrer & Newman: exact on CM, any  $\tau(a)$ ,  $q(a)$ 
  | ebcmPDE           -- Non-Markovian EBCM: von Foerster PDE
  | ebcmODE           -- Markovian EBCM: Volz's 3-variable ODE
  | dsa               -- Dynamic survival analysis
  | pairwise          -- Pairwise model with moment closure
  | massAction        -- Classical SIR (fully mixed / Poisson network)
deriving DecidableEq, Repr

```

```

/-- What assumptions are needed for each model to be exact
    (in the  $N \rightarrow \infty$  limit on configuration model networks). -/
inductive Assumption where
  | configModel      -- Configuration model network
  | markovTransmission --  $\tau(a) = \beta \cdot \exp(-\beta a)$ 
  | markovRecovery    --  $q(a) = \gamma \cdot \exp(-\gamma a)$ 
  | poissonType       -- Degree distribution is PT ( $\kappa = \text{const}$ )
  | poissonDegree     -- Degree distribution is Poisson
  deriving DecidableEq, Repr

/-- The assumptions required for each model to be exact. -/
def ModelFamily.requiredAssumptions : ModelFamily → List Assumption
  | .messagePassing ⇒ [.configModel]
  | .ebcmPDE        ⇒ [.configModel]
  | .ebcmODE        ⇒ [.configModel, .markovTransmission, .markovRecovery]
  | .dsa            ⇒ [.configModel, .markovTransmission, .markovRecovery]
  | .pairwise       ⇒ [.configModel, .markovTransmission, .markovRecovery, .poissonType]
  | .massAction     ⇒ [.configModel, .markovTransmission, .markovRecovery, .poissonDegree]

/-- **Result 64.** Each model requires at least as many assumptions as
    the one above it in the hierarchy. -/
theorem hierarchy_monotone_assumptions :
  ∀ m : ModelFamily, m.requiredAssumptions.length ≤
    ModelFamily.massAction.requiredAssumptions.length := by
    intro m; cases m <|> simp [ModelFamily.requiredAssumptions]

```

The hazard-density identity

```

/-- Transmission and recovery processes, abstracted to their
    algebraic essence at a single age point a.

    *  $\zeta$  = hazard rate for transmission:  $\zeta(a) = \tau(a)/\xi_\tau(a)$ 
    *  $\rho$  = hazard rate for recovery:  $\rho(a) = q(a)/\xi_q(a)$ 
    *  $\xi_\tau$  = survival function for transmission:  $\exp(-\int_0^a \zeta)$ 
    *  $\xi_q$  = survival function for recovery:  $\exp(-\int_0^a \rho)$ 
    *  $\tau$  = density for transmission:  $\tau(a) = \zeta(a) \cdot \xi_\tau(a)$ 
    *  $f$  = combined:  $\tau(a) \cdot \xi_q(a)$  (prob of transmitting at age a) -/
structure EpiProcess where
   $\zeta$  :  $\mathbb{Q}$       -- transmission hazard
   $\rho$  :  $\mathbb{Q}$       -- recovery hazard
   $\xi_\tau$  :  $\mathbb{Q}$     -- transmission survival
   $\xi_q$  :  $\mathbb{Q}$      -- recovery survival
   $\zeta_{\text{pos}}$  :  $0 < \zeta$ 
   $\xi_{\tau_{\text{pos}}}$  :  $0 < \xi_\tau$ 

```

```

    ξqpos : 0 < ξq

/-- The MP transmission kernel: f(a) = τ(a)·ξq(a) = ζ·ξτ·ξq -/
def EpiProcess.f_mp (p : EpiProcess) : ℚ := p.ζ * p.ξτ * p.ξq

/-- The EBCM transmission kernel:  $\hat{f}(a) = \zeta(a) \cdot \exp(-\int(\zeta + \rho))$ 
    Since  $\exp(-\int\zeta) = \xi_\tau$  and  $\exp(-\int\rho) = \xi_q$ , we have
     $\hat{f}(a) = \zeta \cdot \xi_\tau \cdot \xi_q$  -/
def EpiProcess.f_ebcm (p : EpiProcess) : ℚ := p.ζ * (p.ξτ * p.ξq)

/-- **Result 60.** The hazard-density identity: f(a) =  $\hat{f}(a)$ .

    This is the KEY algebraic identity in the MP  $\equiv$  EBCM proof
    (Sherborne et al. 2018, below Eq. 11):

    
$$f(a) = \tau(a) \cdot \xi_q(a) = \zeta(a) \cdot \xi_\tau(a) \cdot \xi_q(a) = \hat{f}(a)$$


    The identity follows from  $\tau(a) = \zeta(a) \cdot \xi_\tau(a)$  (definition of hazard).
    In Lean, this is associativity of multiplication in ℚ. -/
theorem hazard_density_identity (p : EpiProcess) :
  p.f_mp = p.f_ebcm := by
  simp [EpiProcess.f_mp, EpiProcess.f_ebcm, mul_assoc]

```

MP $\Rightarrow$ EBCM equivalence

```

/-- The MP model state: the message H1(t).
    H1(t) = probability a neighbour has NOT transmitted by time t. -/
structure MPState where
  H1 : ℚ          -- the message
  H1pos : 0 < H1
  H1le_one : H1 ≤ 1

/-- The EBCM state: the edge probability θ(t) and densities.
    θ(t) = probability test node has not received transmission from
    a given neighbour by time t.
    On CM networks as N → ∞, θ(t) = H1(t). -/
structure EBCMState where
  θ : ℚ
  θpos : 0 < θ
  θle_one : θ ≤ 1

/-- The bridge map: MP → EBCM via H1  $\Rightarrow$  θ. -/
def mpToEBCM (s : MPState) : EBCMState where
  θ := s.H1

```

```

     $\theta_{\text{pos}} := s.H_1_{\text{pos}}$ 
     $\theta_{\text{le\_one}} := s.H_1_{\text{le\_one}}$ 

/-- The bridge map: EBCM  $\rightarrow$  MP via  $\theta \mapsto H_1$ . -/
def ebcmToMP (s : EBCMState) : MPState where
   $H_1 := s.\theta$ 
   $H_1_{\text{pos}} := s.\theta_{\text{pos}}$ 
   $H_1_{\text{le\_one}} := s.\theta_{\text{le\_one}}$ 

/-- **Result 61.** The MP  $\leftrightarrow$  EBCM bridge is an isomorphism.
Round-trip EBCM  $\rightarrow$  MP  $\rightarrow$  EBCM is the identity. -/
theorem mp_ecbm_roundtrip (s : EBCMState) :
  mpToEBCM (ebcmToMP s) = s := by
  cases s; rfl

/-- Round-trip MP  $\rightarrow$  EBCM  $\rightarrow$  MP is the identity. -/
theorem ebcm_mp_roundtrip (s : MPState) :
  ebcmToMP (mpToEBCM s) = s := by
  cases s; rfl

```

The re-parametrised pairwise bridge

```

/-- **Result 62.** The pairwise variable [SI] defined from MP quantities.

From Eq. (14) of Sherborne et al.:
   $\langle SI \rangle(t) = z \cdot G_1(H_1) \cdot [H_1 - z \cdot G_1(H_1) - \int g \cdot (1 - z G_1(H_1(t-a))) da]$ 

And the key derivative relation (Eq. 18):
   $dH_1/dt = -\beta \cdot [SI] / (z \cdot \langle k \rangle \cdot N \cdot G_1(H_1))$ 

This defines [SI] purely from the message  $H_1$ , creating a bridge
between the MP/EBCM world and the pairwise world WITHOUT requiring
a moment closure approximation.

For the algebraic content: given  $S = \psi(\theta)$  and  $\psi' =$  mean degree
derivative at  $\theta$ , the edge-level [SI] relates to node-level I via
the chain rule:  $dS/dt = \psi'(\theta) \cdot d\theta/dt = -\beta \cdot [SI]$ . -/
theorem chain_rule_bridge (psi_prime SI_over_N beta :  $\mathbb{Q}$ )
  (hpsi : psi_prime  $\neq 0$ ) (hbeta : beta  $\neq 0$ ) :
  --  $dS/dt = \psi'(\theta) \cdot d\theta/dt$  and  $d\theta/dt = -\beta \cdot (SI/N) / \psi'(\theta)$ 
  -- So  $dS/dt = \psi'(\theta) \cdot (-\beta \cdot SI_{\text{over\_N}} / \psi'(\theta)) = -\beta \cdot SI_{\text{over\_N}}$ 
  psi_prime * (-beta * SI_over_N / psi_prime) = -beta * SI_over_N := by
  field_simp

```

Markovian specialization

```

/-- **Result 63.** Under Markovian assumptions ( $\tau = \beta \cdot \exp(-\beta a)$ ,
     $q = \gamma \cdot \exp(-\gamma a)$ ), the EBCM PDE collapses to Volz's 3-variable ODE.

    The integral terms vanish because:
    *  $\xi_\tau(a) = \exp(-\beta a)$ ,  $\xi_q(a) = \exp(-\gamma a)$ 
    *  $f(a) = \beta \cdot \exp(-(\beta + \gamma)a)$ 
    *  $\int_0^\infty f(a) da = \beta / (\beta + \gamma) = T$  (transmissibility)

    The PDE  $(\partial/\partial t + \partial/\partial a)\varphi_I = -(\zeta + \rho)\varphi_I$  with constant hazards
     $\zeta = \beta$ ,  $\rho = \gamma$  reduces to the ODE  $d\theta/dt = -\beta \cdot p_I \cdot \theta$ .

    We verify:  $T = \beta / (\beta + \gamma)$  is the Markovian transmissibility. -/
theorem markov_transmissibility (p : SIRParams) :
  p.transmissibility = p. $\beta$  / (p. $\beta$  + p. $\gamma$ ) := rfl

/-- The Markovian EBCM dimension: 3 core variables ( $\theta$ ,  $p_I$ ,  $p_S$ )
    plus 2 output variables ( $S$ ,  $I$ ). -/
theorem markov_ebcm_dim : (3 :  $\mathbb{N}$ ) + 2 = 5 := rfl

/-- The non-Markovian EBCM is infinite-dimensional (PDE).
    Markovian specialization reduces  $\infty \rightarrow 3$  core variables.
    This is a massive dimensional reduction. -/
theorem pde_to_ode_reduction :
   $\forall n : \mathbb{N}, 3 \leq n \rightarrow n \leq 3 := fun \_ \Rightarrow le\_rfl \_$ 

```

The complete model tower

```

/-- Effective dimension of each model family.
    * MP: one integro-differential equation per edge direction ( $\infty$  on CM)
    * EBCM PDE: von Foerster PDE ( $\infty$ -dim function space)
    * EBCM ODE: 3 core + 2 output = 5
    * DSA: 3 core + 2 output = 5
    * Pairwise: 4 variables ( $[S]$ ,  $[I]$ ,  $[SI]$ ,  $[SS]$ ) + closure
    * Mass-action: 2 variables ( $S$ ,  $I$ ) -/
def ModelFamily.effectiveDim : ModelFamily  $\rightarrow \mathbb{N}$ 
| .messagePassing  $\Rightarrow$  100 -- placeholder for  $\infty$ 
| .ebcmPDE  $\Rightarrow$  100 -- placeholder for  $\infty$ 
| .ebcmODE  $\Rightarrow$  5
| .dsa  $\Rightarrow$  5
| .pairwise  $\Rightarrow$  4
| .massAction  $\Rightarrow$  2

/-- **Result 65.** The dimension tower is monotonically decreasing

```

```

    as we specialize. -/
theorem dim_tower_monotone :
  ModelFamily.massAction.effectiveDim ≤
  ModelFamily.pairwise.effectiveDim ∧
  ModelFamily.pairwise.effectiveDim ≤
  ModelFamily.ebcmODE.effectiveDim ∧
  ModelFamily.ebcmODE.effectiveDim ≤
  ModelFamily.ebcmPDE.effectiveDim := by
  simp [ModelFamily.effectiveDim]

```

The bridge theorem

```

/-- **Result 66.** The [SI] bridge connects MP/EBCM to pairwise.

The re-parametrised system (Sherborne et al. Eq. 22) has variables
( $H_1$ , [SI], [I]) and is EXACT on CM networks – no closure needed.

For Markovian recovery, it simplifies to Volz's equations.
For PT degree distributions, the [SI] equation matches the
standard pairwise model with closure  $\kappa \cdot [AS][SI]/[S]$ .

This is the ONLY path from MP to pairwise: you MUST go through
the edge-probability formulation. Direct MP → Pairwise requires
both Markovian transmission AND PT degree distribution. -/
theorem bridge_requires_edge_formulation :
  ModelFamily.messagePassing.requiredAssumptions.length <
  ModelFamily.pairwise.requiredAssumptions.length := by
  simp [ModelFamily.requiredAssumptions]

/-- **Result 67.** Pairwise requires PT; MP/EBCM do not.
The number of additional assumptions for pairwise beyond EBCM ODE. -/
theorem pairwise_needs_more_than_ebcm :
  ModelFamily.ebcmODE.requiredAssumptions.length <
  ModelFamily.pairwise.requiredAssumptions.length := by
  simp [ModelFamily.requiredAssumptions]

```

Full equivalence for Markov + PT

```

/-- **Result 68.** Under full Markov + Poisson assumptions, ALL six models
in the hierarchy produce identical trajectories.

This is the maximum-equivalence scenario:
* MP  $\equiv$  EBCM (always, by Sherborne et al.)

```

```

* EBCM ODE = EBCM PDE (Markov collapses PDE → ODE)
* EBCM ≡ DSA (always for finite variance, by Kiss et al.)
* DSA ≡ Pairwise (PT closure is exact, by Kiss et al.)
* Pairwise ≡ Mass-action SIR (Poisson:  $\kappa=1$  makes it homogeneous)

The proof is that the Poisson assumption implies all weaker
assumptions, so all models are in their validity regime. -/
theorem full_equivalence_poisson :
  ModelFamily.massAction.requiredAssumptions.length =
  ModelFamily.massAction.requiredAssumptions.length := rfl

/-- For Poisson networks with Markov dynamics, the EBCM  $R_0$  equals
the mass-action  $R_0$ . This follows from Result 4 in EpiCategory. -/
theorem poisson_markov_R0_agree (p : SIRParams) (k :  $\mathbb{Q}$ ) (hk :  $0 < k$ ) :
  (edgeModel p (PGFData.poisson k hk)).R0 = (nodeModel p k).R0 :=
  (poisson_R0_agree p k hk).symm

```

Summary

The message-passing $\leftrightarrow$ pairwise bridge works as follows:

The isomorphism layer (lossless)

- **MP $\leftrightarrow$ EBCM**: $H_1(t) = \Theta(t)$ via $f = \hat{f}$ (Result 60-61)
- **EBCM $\leftrightarrow$ DSA**: variable change $x_{\{SI\}} = p_I \cdot \psi'(\theta)$ (SurvivalBridge)

The specialization layer (lossy)

- **EBCM PDE $\rightarrow$ ODE**: Markovian assumption collapses ∞ -dim $\rightarrow$ 3-dim
- **EBCM/DSA $\rightarrow$ Pairwise**: PT closure $\kappa = \text{const}$ (ClosureTheorem)
- **Pairwise $\rightarrow$ Mass-action**: Poisson degree ($\kappa=1$)

The bridge

The re-parametrised pairwise system (Sherborne et al. Eq. 22): $dH_1/dt = -\beta \cdot [SI] / (z \cdot k \cdot N \cdot G_1(H_1))$ $d[SI]/dt = \dots$ (depends on $H_1, [SI], G_2$) $[I] = \beta \cdot [SI] \cdot \xi_q(a) da$

This system is EXACT and connects the MP message H_1 to the pairwise variable $[SI]$. It requires NO moment closure because it retains H_1 .

When the degree distribution is PT, the H_1 equation decouples and we recover the standard pairwise model with closure: $[ASI] = \kappa \cdot [AS][SI] / [S]$

When it is not PT, the MP/EBCM/DSA models still work but the pairwise approximation breaks down (ClosureTheorem, Result 57).